

MLPerf Storage v3.0

System Description - StorMeshAI

Submitter	HolmesAI Limited	Division	CLOSED
System name	StorMeshAI	Availability	Available

StorMeshAI is designed as a shared remote parallel filesystem for scalable, high-throughput AI training data access over InfiniBand/RDMA.

1. Solution overview

Deployment	HolmesAI Limited cloud service; region and availability zone are not disclosed.
Architecture	Shared remote parallel filesystem with POSIX/file access through the proprietary StorMeshAI client.
Storage scale	Three storage nodes, each with nine 4.76 TiB NVMe devices in a distributed chunk pool.
Capacity	128.52 TiB raw; 42.84 TiB usable after three-way replication.
Data fabric	The client and each of the three storage nodes have two independent 400 Gb/s NDR InfiniBand links.
Service network	50 Gb/s Ethernet bond for management, control-plane, and filesystem metadata traffic.

2. Client infrastructure

- **Quantity and platform:** 1 x Nettrix R620 G50.
- **Processors:** 2 x Intel Xeon Platinum 8458P; 88 physical cores and 176 logical CPUs total.
- **Memory:** 1 TiB DDR5-4800 (16 x 64 GB); 1007 GiB visible to the operating system.
- **Operating system:** Ubuntu 24.04.3 LTS, Linux kernel 6.8.0-117-generic, x86_64.
- **Data network:** 2 x independent 400 Gb/s NDR InfiniBand/RDMA links; both were Active and LinkUp when inspected.
- **Service network:** 1 x 50 Gb/s Ethernet bond formed from two 25 Gb/s interfaces.
- **Client execution:** 32 I/O threads, with work scheduled across both NUMA node 0 and NUMA node 1; no other benchmark-specific client tuning is disclosed.

3. Storage infrastructure

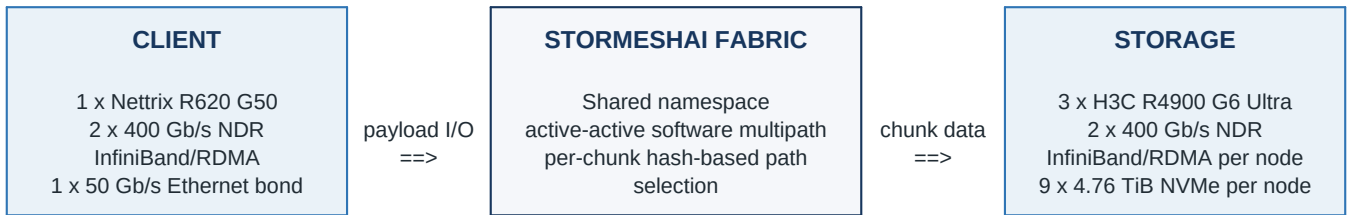
- **Quantity and platform:** 3 x H3C UniServer R4900 G6 Ultra storage nodes.
- **Processors per node:** 2 x Intel Xeon Gold 6530; 64 physical cores and 128 logical CPUs total.
- **Memory per node:** 1 TiB nominal; 1007 GiB visible to the operating system.
- **Operating system:** CentOS Linux 7 (Core), Linux kernel 5.10.0-20.el7, x86_64.
- **Data devices per node:** 9 x 4.76 TiB NVMe. Vendor, model, and media type are not disclosed.
- **Data organization:** the NVMe devices participate directly in the distributed StorMeshAI chunk pool; they are not presented as a storage-node-local filesystem.
- **Services:** StorMeshAI metadata, data, management, and control-plane services.

Raw capacity: 3 nodes x 9 devices x 4.76 TiB = 128.52 TiB

Usable capacity: 128.52 TiB / 3 replicas = 42.84 TiB

4. System and network topology

The client mounts the shared filesystem exposed by StorMeshAI. The client and each of the three storage nodes have two independent 400 Gb/s NDR InfiniBand links. All eight links were Active and LinkUp when inspected. Management, control-plane, service discovery, heartbeat, and filesystem metadata operations use the Ethernet service network.



Observed link inventory

Node	InfiniBand devices	Observed state
Client	mlx5_2, mlx5_3	Each 400 Gb/s NDR; Active and LinkUp
Storage node 1	mlx5_0, mlx5_1	Each 400 Gb/s NDR; Active and LinkUp
Storage node 2	mlx5_0, mlx5_1	Each 400 Gb/s NDR; Active and LinkUp
Storage node 3	mlx5_0, mlx5_1	Each 400 Gb/s NDR; Active and LinkUp

On the client, bond0 is an IEEE 802.3ad LACP bond comprising two 25 Gb/s Ethernet interfaces. Both slave links were Up, and ethtool reported a 50 Gb/s aggregate bond speed.

Traffic class	Network path	Behavior
File payload	2 x 400 Gb/s NDR InfiniBand/RDMA	StorMeshAI is configured to distribute file-payload traffic across the two links using per-chunk hash-based path selection.
Filesystem metadata	50 Gb/s Ethernet bond	Operations including <i>readdir</i> and <i>getattr</i> use the Ethernet path.
Management and control	50 Gb/s Ethernet bond	Service discovery, heartbeat, management, and control-plane TCP traffic.

The InfiniBand switch model is not disclosed; the deployment uses a 400 Gb/s NDR fabric.

5. Software stack and configuration

Component	Description
Storage software	StorMeshAI remote parallel filesystem; product version not disclosed.
Client software	Proprietary StorMeshAI POSIX/file client; client version not disclosed.
Client operating system	Ubuntu 24.04.3 LTS, Linux kernel 6.8.0-117-generic, x86_64.
Storage layout	Distributed chunk pool with three-way replication and per-chunk path selection.
Client I/O	32 I/O threads using both NUMA nodes. No additional benchmark-specific tuning is disclosed.

6. Disclosure notes

- Region, availability zone, product software version, client software version, NVMe vendor/model/media type, and InfiniBand switch model are not disclosed.
- Benchmark workload parameters and reported results are intentionally not repeated here; the repository result artifacts and generated results tables remain authoritative.
- Storage-node information is documented in this PDF because the system-description YAML schema used by this submission does not include a storage-server block.